



UNIVERSITY OF
MICHIGAN

EECS 453

Principles of Machine Learning

Lecture 3 – Linear Regression

Instructor: Prof. Qing Qu

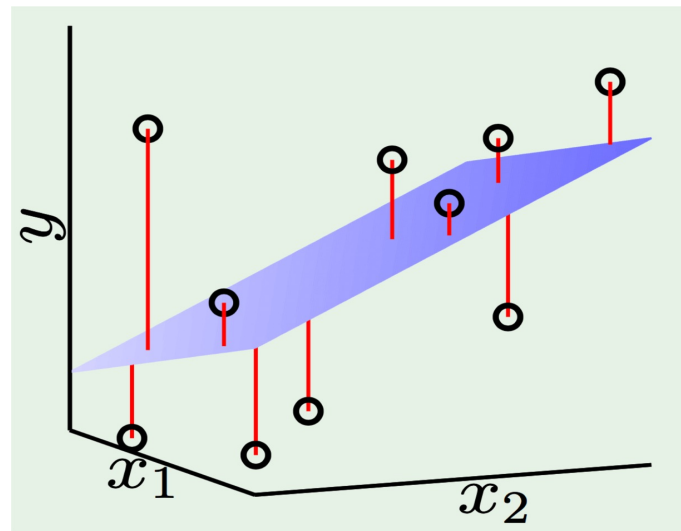
Lecture Agenda

- **More on Linear Regression**
 - Maximum Likelihood Estimation ✓
 - Dealing with Nonlinearity ✓
 - Ridge Regression/Tikhonov Regularization }
 - Feature Selection and LASSO

Linear Regression Problem

- **Data:** given past sales data $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$, where $\mathbf{x}^{(i)} \in \mathbb{R}^d$ represents living size, lot size, # floors, conditions, schools ...
 $y^{(i)}$ represents sales price;
- **Hypothesis/model:**
 $y^{(i)} \approx \mathbf{x}^{(i)\top} \boldsymbol{\theta}$, where $\boldsymbol{\theta} \in \mathbb{R}^d$
- **Learning problem:** solve a *least squares* (LS) problem

$$\boldsymbol{\theta}_\star = \arg \min_{\boldsymbol{\theta}} \frac{1}{n} \sum_{i=1}^n (y^{(i)} - \mathbf{x}^{(i)\top} \boldsymbol{\theta})^2$$



Solution of Least Squares

- Objective:

$$\begin{aligned} L(\boldsymbol{\theta}) &= \underbrace{\|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2}_{\text{}} = (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y}) \\ &= \boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y} \end{aligned}$$

- Take the gradient:

$$\underbrace{\nabla_{\boldsymbol{\theta}} L(\boldsymbol{\theta})}_{\text{}} = 2\mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

- Set the gradient to zero: the optimal solution $\boldsymbol{\theta}_\star$ satisfies

$$\underbrace{\mathbf{X}^\top \mathbf{X} \boldsymbol{\theta}_\star}_{\text{}} = \mathbf{X}^\top \mathbf{y}$$

- Question: how to solve for $\boldsymbol{\theta}_\star$ based on the above linear equations?

Solution of Least Squares

Case I: $\mathbf{X} \in \mathbb{R}^{n \times d}$ has **full** column rank ($n \geq d$)

- $\mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{d \times d}$ is *non-singular* (i.e., invertible)
- The optimal solution has a *closed-form*

$$\boldsymbol{\theta}_\star = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^\dagger \mathbf{y}$$

where

$$\mathbf{X}^\dagger = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \in \mathbb{R}^{d \times n}$$

denotes the **pseudo-inverse** of the matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, whenever it has full column rank.

Note: We have $\mathbf{X}^\dagger \mathbf{X} = \mathbf{I}$, but not necessarily $\mathbf{X} \mathbf{X}^\dagger = \mathbf{I}$.

Solution of Least Squares

Case II: $\mathbf{X} \in \mathbb{R}^{n \times d}$ does NOT have full column rank

- $\boldsymbol{\theta}$ does not have unique solutions
- Typically, $n < d$, thus
“overfitting”: more model parameters than can be justified by the number of data points
- How to obtain unique and meaningful solution?

“regularization”: e.g., $\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2$

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- More on Linear Regression

- **Maximum Likelihood Estimation** (MLE)
- Dealing with Nonlinearity
- Ridge Regression/Tikhonov Regularization
- Feature Selection and LASSO

Probability Example: Flip Coin

- **Setup:** flip a coin with Head (H) and Back (B)
- **Outcome:** $\mathcal{D} = \{\underline{H}, \underline{H}, \underline{B}, \underline{H}, \underline{B}\}$, $\underline{k}$ heads out of $\underline{n}$ flips.

Probability Example: Flip Coin

- **Setup:** flip a coin with Head (H) and Back (B)
- **Outcome:** $\mathcal{D} = \{H, H, B, H, B\}$, k heads out of n flips.
- **Assumption:**
 - Suppose $\Pr[\text{Head}] = \theta$ $\Pr[\text{Back}] = 1 - \theta$
 - All flips are i.i.d., $(i+1)$ -th flip is independent of i -th outcome

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- **Question:** 

What is the probability it will be head?

Conditional Probability: Flip Coin

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 - Suppose $\Pr[\text{Head}] = \theta$ $\Pr[\text{Back}] = 1 - \theta$
 - All flips are i.i.d., $(i+1)$ -th flip is independent of i -th outcome
- **Question:**

What is the probability it will be head?

- **Task:** estimate θ from the observed data
(using maximum likelihood estimation (MLE))

Conditional Probability

- For two given events $\mathcal{A}$ and $\mathcal{B}$:

$$\Pr[\underline{\mathcal{A}} \mid \underline{\mathcal{B}}] := \frac{\Pr[\mathcal{A} \cap \mathcal{B}]}{\Pr[\mathcal{B}]}$$

- We care about the likelihood of the outcome for coin flip:

$\Pr[\mathcal{D} \mid \theta]$: the probability of the outcome $\mathcal{D}$ $\in \{H, B\}$ given parameter θ

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- We care about the likelihood of the outcome for coin flip:

$\Pr[\mathcal{D} \mid \theta]$: the probability of the outcome $\mathcal{D} \in \{H, B\}$ given parameter θ

- Write out the likelihood:**

$$\Pr[\mathcal{D} \mid \theta] = \theta^k (1 - \theta)^{n-k}$$

$$\max_{\theta} \theta^k (1 - \theta)^{n-k}$$

Since k heads out of n flips for $\mathcal{D} = \{H, H, B, H, B\}$

Maximum Likelihood Estimation (MLE)

- The principle of MLE:

Maximize the probability $\Pr[\mathcal{D} \mid \theta]$ of the observed data over the parameter θ

$$\begin{aligned}\hat{\theta}_{MLE} &= \arg \max_{\theta} \Pr[\mathcal{D} \mid \theta] \\ &= \arg \max_{\theta} \theta^k (1 - \theta)^{n-k} = \mathcal{L}(\theta) \\ \log \mathcal{L}(\theta) &= k \log \theta + (n-k) \log (1 - \theta)\end{aligned}$$

Maximum Likelihood Estimation (MLE)

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Maximize the probability $\Pr[\mathcal{D} \mid \theta]$ of the observed data over the parameter θ

$$\hat{\theta}_{MLE} = \arg \max_{\theta} \Pr[\mathcal{D} \mid \theta]$$

$$= \arg \max_{\theta} \theta^k (1 - \theta)^{n-k}$$

- Log-likelihood:

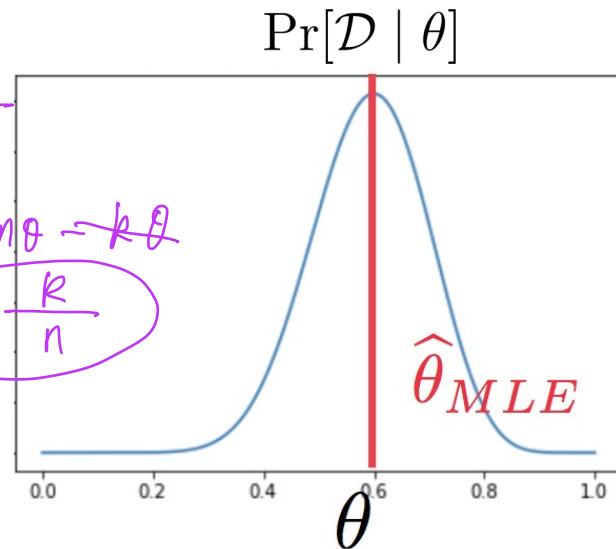
$$\frac{\partial}{\partial \theta} \log(\Pr[\mathcal{D} \mid \theta]) = k \log(\theta) + (n - k) \log(1 - \theta)$$

$$\frac{\partial \log(x)}{\partial x} = \log'(x) = \frac{1}{x}$$

$$\frac{k}{\theta} = \frac{n-k}{1-\theta}$$

$$\Leftrightarrow k - k\theta = n\theta - k\theta$$

$$\Leftrightarrow \theta = \frac{k}{n}$$



Maximum Likelihood Estimation (MLE)

- Why log likelihood?

- The log function is monotonic increasing, thus

$$\arg \max_{\theta} \Pr[\mathcal{D} \mid \theta] \equiv \arg \max_{\theta} \log (\Pr[\mathcal{D} \mid \theta])$$

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- It is *extremely* useful for joint likelihood of i.i.d. points $\mathcal{D} = \{x^{(1)}, \dots, x^{(n)}\}$

$$\underbrace{p(\mathcal{D} \mid \theta)} = \prod_{i=1}^n p(x^{(i)} \mid \theta) \quad \longrightarrow \quad \underbrace{\log (p(\mathcal{D} \mid \theta))} = \sum_{i=1}^n \log \left(\underbrace{p(x^{(i)} \mid \theta)} \right)$$

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- Log-likelihood:** $\mathcal{L}(\theta) = \log (\Pr[\mathcal{D} \mid \theta]) = k \log (\theta) + (n - k) \log (1 - \theta)$

$$\mathcal{L}'(\theta) = \frac{\partial \mathcal{L}(\theta)}{\partial \theta} = \frac{k}{\theta} - \frac{n - k}{1 - \theta} \quad \mathcal{L}'(\theta) = 0$$

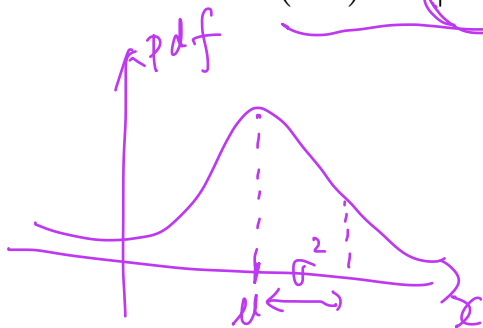
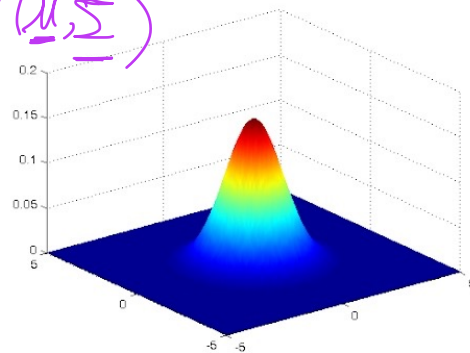
$$\hat{\theta}_{MLE} = \frac{k}{n}$$

Continuous Variable? MLE for Gaussian

- **Multi-variate Gaussian distribution:** we say $\mathbf{x} \in \mathbb{R}^d$ follows Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ (positive definite)

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

$\mathbf{x} = \{x_1, \dots, x_n\}$

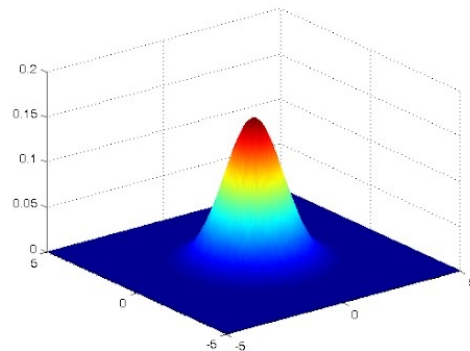


$$x \sim \mathcal{N}(\mu, \sigma^2) \quad \text{pdf}(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x-\mu)^2}{2\sigma^2} \right)$$

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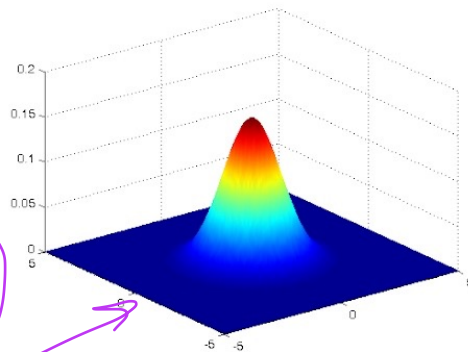


- **Data:** draw a set of i.i.d. samples $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ follow $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$

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$$\mathcal{N}(\mathbf{x} \mid \underbrace{\boldsymbol{\mu}, \boldsymbol{\Sigma}}_{\boldsymbol{\theta}}) = \frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$



- **Data:** draw a set of i.i.d. samples $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ follow $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$
 - **Question:** what is the $\hat{\boldsymbol{\theta}}_{MLE}$ for $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\Sigma})$, given samples $\mathcal{D}$?
- $\hat{\boldsymbol{\mu}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)}$

MLE for Gaussian

- **Likelihood:** set parameters $\theta = (\mu, \Sigma)$

$$p(\underline{\mathcal{D}} | \underline{\theta}) = \prod_{i=1}^n p(\underline{x}^{(i)} | \underline{\theta})$$

$$p(x^{(1)}, x^{(2)}, \dots, x^{(n)} | \theta) = \prod_{i=1}^n p(x^{(i)} | \theta)$$

$$= \frac{1}{(2\pi)^{nd/2} |\Sigma|^{n/2}} \prod_{i=1}^n \exp \left(-\frac{1}{2} (\underline{x}^{(i)} - \mu)^\top \Sigma^{-1} (\underline{x}^{(i)} - \mu) \right)$$

MLE for Gaussian

- **Likelihood:** set parameters $\theta = (\mu, \Sigma)$

$$\log p(\mathcal{D} | \theta) = \prod_{i=1}^n p(\mathbf{x}^{(i)} | \theta)$$

$$\log \exp(x) = x$$

$$= \frac{1}{(2\pi)^{nd/2} |\Sigma|^{n/2}} \prod_{i=1}^n \exp \left(-\frac{1}{2} (\mathbf{x}^{(i)} - \mu)^\top \Sigma^{-1} (\mathbf{x}^{(i)} - \mu) \right)$$

- **Log-likelihood:** set parameters $\theta = (\mu, \Sigma)$

$$\nabla_{\mu} \mathcal{L}(\theta) = 0$$

$$\nabla_{\Sigma^{-1}} \mathcal{L}(\theta) = 0$$

$$\nabla \mathcal{L}(\theta) = \sum_{i=1}^n \log(p(\mathbf{x}^{(i)} | \theta))$$

$$= -\frac{nd}{2} \log(2\pi) - \frac{n}{2} \log(|\Sigma|) - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}^{(i)} - \mu)^\top \Sigma^{-1} (\mathbf{x}^{(i)} - \mu)$$

MLE for Gaussian

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

- MLE for the mean: compute the derivative of $\mathcal{L}(\theta)$ with respect to μ

$$\nabla_{\mu} \mathcal{L}(\theta) = \nabla_{\mu} \left(\frac{1}{2} \sum_{i=1}^n (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu) \right)$$

$$\nabla_{\mu} \mathcal{L}(\theta) = -\Sigma^{-1} \sum_{i=1}^n (x^{(i)} - \mu) = 0$$

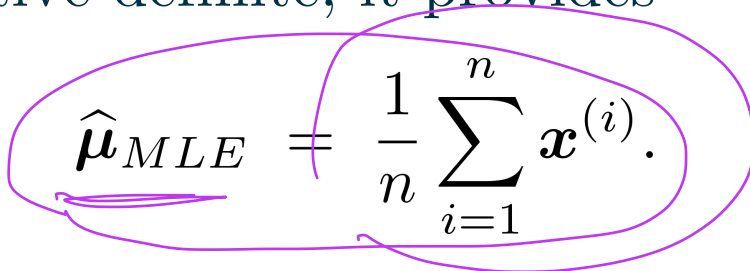
$$\Leftrightarrow \sum_{i=1}^n (x^{(i)} - \mu) = 0 \quad \Leftrightarrow \mu = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

MLE for Gaussian

- **MLE for the mean:** compute the derivative of $\mathcal{L}(\theta)$ with respect to μ

$$\nabla_{\mu} \mathcal{L}(\theta) = \Sigma^{-1} \sum_{i=1}^n \left(\mathbf{x}^{(i)} - \mu \right)$$

Setting the gradient $\nabla_{\mu} \mathcal{L}(\theta) = \mathbf{0}$, together with the fact that Σ^{-1} is positive definite, it provides


$$\hat{\mu}_{MLE} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}^{(i)}.$$

MLE for Gaussian

- **MLE for the variance:** given the MLE estimate of the mean, compute the derivative of $\mathcal{L}(\boldsymbol{\theta})$ with respect to $\boldsymbol{\Sigma}^{-1}$

$$\nabla_{\boldsymbol{\Sigma}^{-1}} \mathcal{L}(\boldsymbol{\theta}) = \frac{n}{2} \boldsymbol{\Sigma} - \frac{1}{2} \sum_{i=1}^n \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right) \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right)^{\top}$$

MLE for Gaussian

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Setting the gradient $\nabla_{\boldsymbol{\Sigma}^{-1}} \mathcal{L}(\boldsymbol{\theta}) = \mathbf{0}$, it provides

$$\hat{\boldsymbol{\Sigma}}_{MLE} = \frac{1}{n} \sum_{i=1}^n \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right) \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right)^{\top}.$$

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- **MLE for the variance:** given the MLE estimate of the mean, compute the derivative of $\mathcal{L}(\boldsymbol{\theta})$ with respect to $\boldsymbol{\Sigma}^{-1}$

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$$\hat{\boldsymbol{\Sigma}}_{MLE} = \frac{1}{n} \sum_{i=1}^n \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right) \left(\mathbf{x}^{(i)} - \hat{\boldsymbol{\mu}}_{MLE} \right)^{\top}.$$

Fact: $\nabla \log(|\mathbf{A}|) = \mathbf{A}^{-\top}$ for any invertible $\mathbf{A}$.

Unbiased and Biased Estimators

Suppose $\mathbf{x}^{(i)} \sim_{i.i.d.} \mathcal{N}(\boldsymbol{\mu}_\star, \boldsymbol{\Sigma}_\star)$

- MLE for the mean is unbiased:

$$\mathbb{E}[\hat{\boldsymbol{\mu}}_{MLE}] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbf{x}^{(i)}] = \boldsymbol{\mu}_\star$$

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- MLE for the variance is biased:

$$\mathbb{E}[\hat{\boldsymbol{\Sigma}}_{MLE}] \neq \boldsymbol{\Sigma}_\star$$

$\frac{n-1}{n} \boldsymbol{\Sigma}_\star$ $n \rightarrow +\infty$ $\frac{n-1}{n} \rightarrow 1$

Maximum Likelihood Estimation: Review

- Observe data $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ following $p(\mathbf{x} \mid \theta_*)$
 $\theta = (\mu, \Sigma)$

Maximum Likelihood Estimation: Review

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- Build likelihood function $p(\mathcal{D} \mid \boldsymbol{\theta}) = \prod_{i=1}^n p(\mathbf{x}^{(i)} \mid \boldsymbol{\theta})$ for some parameter $\boldsymbol{\theta}$

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- Log-likelihood: $\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \log(p(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}))$

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- Maximum likelihood estimator: $\hat{\boldsymbol{\theta}}_{MLE} = \arg \max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$

Maximum Likelihood Estimation: Review

- Observe data $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\}$ following $p(\mathbf{x} | \boldsymbol{\theta}_*)$
 $f(\boldsymbol{\theta}) = p_1(\boldsymbol{\theta}) (p_2(\boldsymbol{\theta}) p_3(\boldsymbol{\theta})) + p_1(\boldsymbol{\theta}) (p_2(\boldsymbol{\theta})) p_3(\boldsymbol{\theta})$
- Build likelihood function $p(\mathcal{D} | \boldsymbol{\theta}) = \prod_{i=1}^n p(\mathbf{x}^{(i)} | \boldsymbol{\theta})$ for some parameter $\boldsymbol{\theta}$
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- Log-likelihood: $\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^n \log(p(\mathbf{x}^{(i)} | \boldsymbol{\theta}))$
- Maximum likelihood estimator: $\hat{\boldsymbol{\theta}}_{MLE} = \arg \max_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$

Key point: know the prior distribution $p(\mathbf{x} | \boldsymbol{\theta})$

LS for Linear Regression: MLE Interpretation

- Recap Least Squares (LS): $\min_{\boldsymbol{\theta}} \frac{1}{n} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$

$$y_i \approx \mathbf{x}_i^\top \boldsymbol{\theta}$$

- Recap our linear model: $y^{(i)} \approx \mathbf{x}^{(i)\top} \boldsymbol{\theta}$

- Consider the following probability model:

$$y_i = \mathbf{x}^{(i)\top} \boldsymbol{\theta} + \epsilon^{(i)}$$

with

$$\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2)$$

where $\epsilon^{(i)}$ are i.i.d. for $i = 1, \dots, n$

- In matrix notation

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\epsilon} \quad \text{with} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

with $\boldsymbol{\Sigma} = \text{diag}(\sigma^2, \dots, \sigma^2)$.



$$\boldsymbol{\Sigma} = \sigma^2 \mathbf{I} = \begin{pmatrix} \sigma^2 & & \\ & \ddots & \\ & & \sigma^2 \end{pmatrix}$$

LS for Linear Regression: MLE Interpretation

Equivalently,

$$\underline{\epsilon} = \underline{y} - \underline{X\theta} \sim \mathcal{N}(\mathbf{0}, \Sigma) \quad \Leftrightarrow \quad \underline{y} = \underline{X\theta} + \underline{\epsilon}$$
$$y \sim \mathcal{N}(\underline{x\theta}, \Sigma)$$

LS for Linear Regression: MLE Interpretation

Equivalently,

$$\epsilon = \mathbf{y} - \mathbf{X}\underline{\boldsymbol{\theta}} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

$$\mathcal{D} = \{y_1, \dots, y_n\}$$

The likelihood: the probability of observed data given parameters $\boldsymbol{\theta}$

$$\begin{aligned} \underline{p(\mathcal{D} \mid \boldsymbol{\theta})} &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} \boldsymbol{\epsilon}^\top \boldsymbol{\Sigma}^{-1} \boldsymbol{\epsilon} \right) \\ &= \frac{1}{(2\pi)^{n/2}} \frac{1}{|\boldsymbol{\Sigma}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \right) \end{aligned}$$

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Handwritten notes: $\boldsymbol{\Sigma}^{-1} = \begin{pmatrix} \sigma^{-2} & & \\ & \ddots & \\ & & \sigma^{-2} \end{pmatrix}$ and $\sigma^{-2} \mathbf{I}$

Log-likelihood:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) &= \log(p(\mathcal{D} \mid \boldsymbol{\theta})) = \text{Constant} - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\theta}) \\ &= \text{Constant} - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \end{aligned}$$

Handwritten notes: $\rightarrow \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2$

LS for Linear Regression: MLE Interpretation

MLE principle:

$$\begin{aligned}\hat{\boldsymbol{\theta}}_{MLE} &= \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \\ &= \mathbf{X}^\dagger \mathbf{y}\end{aligned}$$

X full column rank

$$\mathbf{X}^\dagger = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}$$

LS for Linear Regression: MLE Interpretation

MLE principle:

$$\begin{aligned}\hat{\boldsymbol{\theta}}_{MLE} &= \underset{\boldsymbol{\theta}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \\ &= \mathbf{X}^\dagger \mathbf{y}\end{aligned}$$

The least squares is maximum likelihood estimator under Gaussian noise

How Good is MLE?

- Generative model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\theta}_\star + \boldsymbol{\epsilon} \quad \text{with} \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$$

- Goal: to learn the target $\boldsymbol{\theta}_\star$

- MLE or LS solution:
$$\begin{aligned}\hat{\boldsymbol{\theta}}_{MLE} &= \mathbf{X}^\dagger \mathbf{y} = \mathbf{X}^\dagger (\mathbf{X}\boldsymbol{\theta}_\star + \boldsymbol{\epsilon}) \\ &= \underbrace{\mathbf{X}^\dagger \mathbf{X}}_{\mathbf{I}} \boldsymbol{\theta}_\star + \mathbf{X}^\dagger \boldsymbol{\epsilon} \\ &= \boldsymbol{\theta}_\star + \mathbf{X}^\dagger \boldsymbol{\epsilon}\end{aligned}$$

- Unbiased:
$$\mathbb{E}[\hat{\boldsymbol{\theta}}_{MLE}] = \boldsymbol{\theta}_\star$$

- Variance:
$$\text{Cov}(\hat{\boldsymbol{\theta}}_{MLE}) = (\mathbf{X}^\top \mathbf{X})^{-1} \text{ (exercise)}$$

Lecture Agenda

- More on Linear Regression
 - Maximum Likelihood Estimation
 - **Dealing with Nonlinearity**
 - Ridge Regression/Tikhonov Regularization
 - Feature Selection and LASSO

Dealing with Nonlinear Data

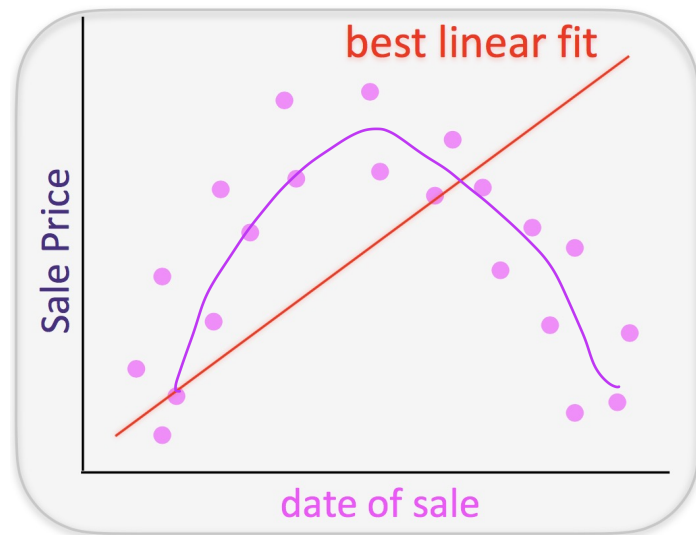
- **Data:** given past sales data $\{(\underline{x}^{(i)}, \underline{y}^{(i)})\}_{i=1}^n$, where $\underline{x}^{(i)}$ represents {date of sale, square feet, ...}, and $\underline{y}^{(i)}$ represents sales price

- Model:

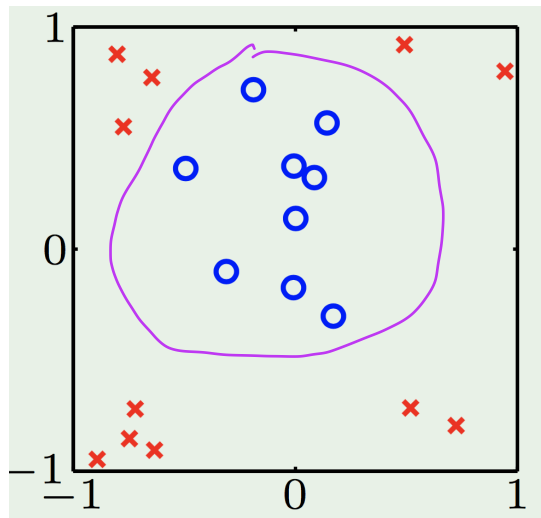
$$\underline{y} \approx \underline{X} \underline{\theta}_\star$$

- Learning problem:

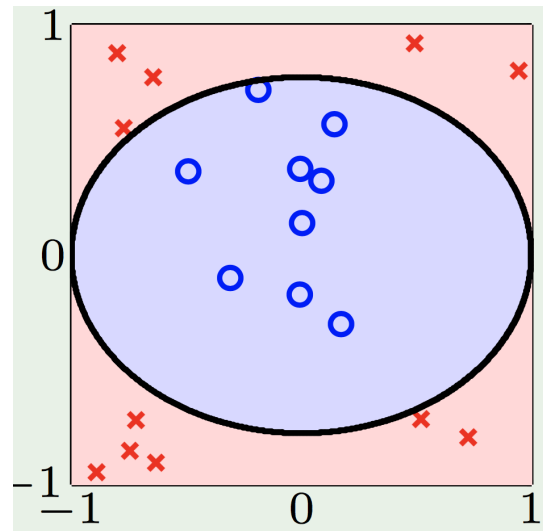
$$\begin{aligned} \underline{\theta}_\star &= \arg \min_{\underline{\theta}} \|\underline{y} - \underline{X} \underline{\theta}\|_2^2 \\ &= \underline{X}^\dagger \underline{y}. \end{aligned}$$



Dealing with Nonlinear Data



Data

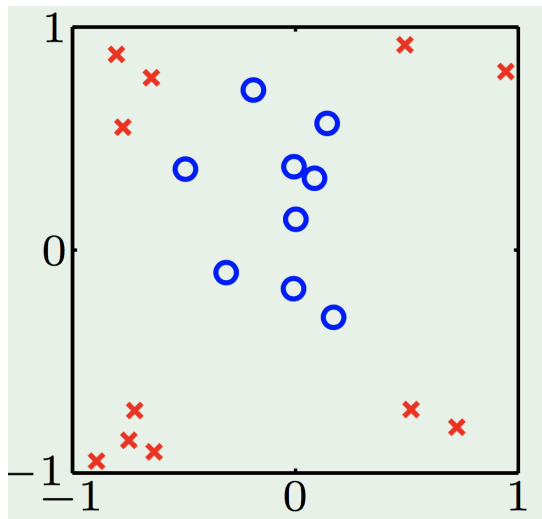


Nonlinear Model

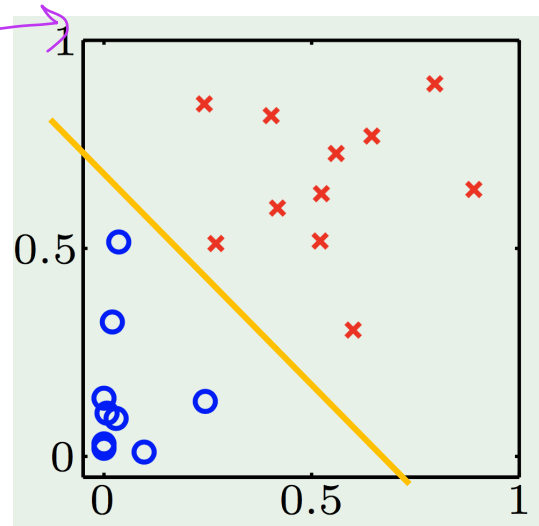
One possibility: nonlinear model?

Transform the Data Nonlinearly

$$(\underline{x_1, x_2}) \mapsto (\underline{x_1^2, x_2^2})$$



Data



Nonlinear Model

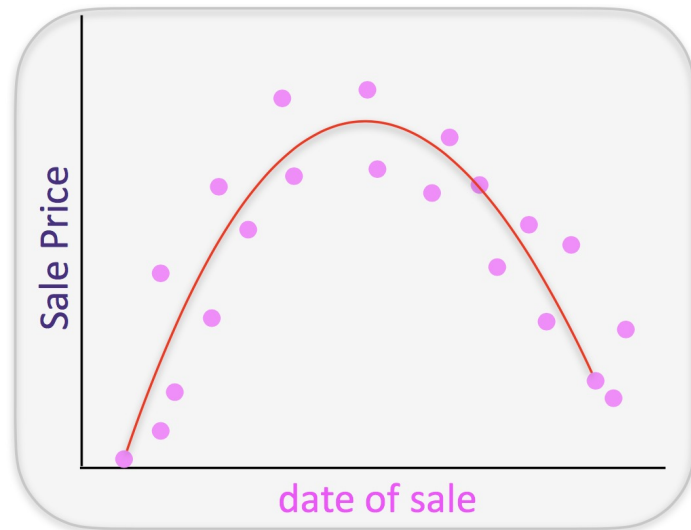
Generalized Linear Regression

- **Nonlinear transformation**

$$\Phi(\mathbf{x}) : \mathbb{R}^d \mapsto \mathbb{R}^p$$

- **Example with** $d = 1$

$$\Phi(\mathbf{x}) = \begin{bmatrix} x \\ x^2 \\ \vdots \\ x^p \end{bmatrix}$$



Generalized Linear Regression

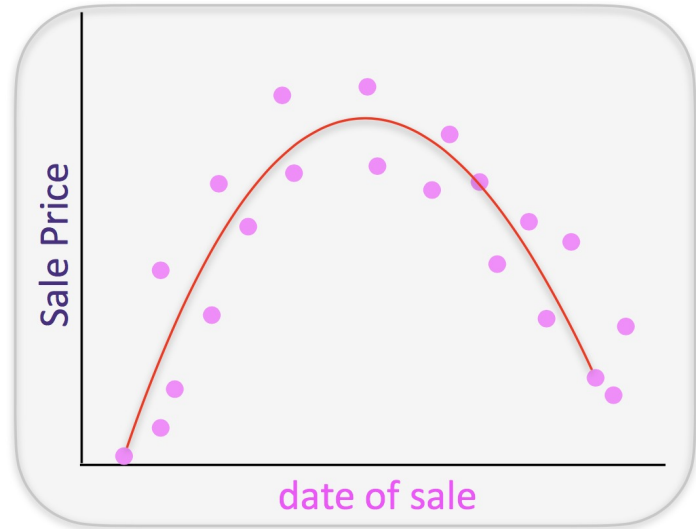
- Original training data:

$$\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$$

- Transformed training data:

$$\{(\Phi(\mathbf{x}^{(i)}), y^{(i)})\}_{i=1}^n$$

$$x \rightarrow \Phi(x)$$



Generalized Linear Regression

- Original training data:

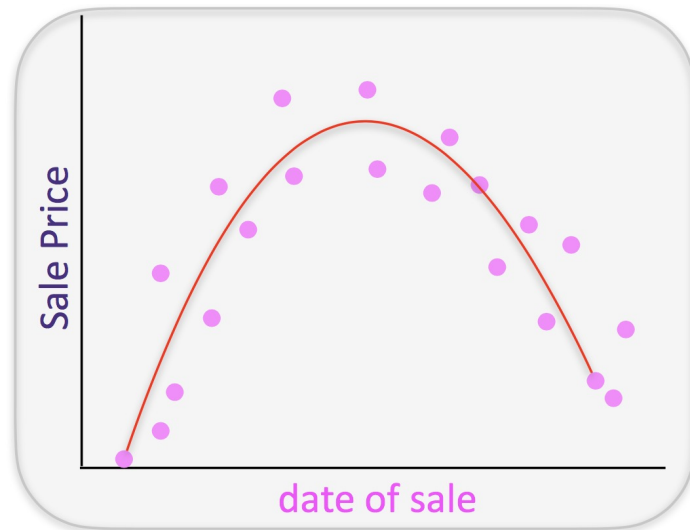
$$\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$$

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$$\{(\Phi(\mathbf{x}^{(i)}), y^{(i)})\}_{i=1}^n$$

- Model: linear in $\boldsymbol{\theta}$

$$y^{(i)} \approx \underbrace{\Phi(\mathbf{x}^{(i)})^\top \boldsymbol{\theta}}$$



Generalized Linear Regression

- **Original training data:**

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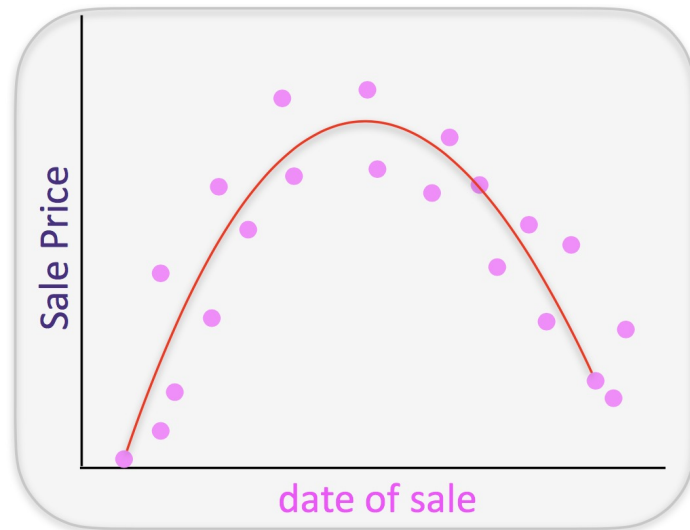
$$\{(\Phi(\mathbf{x}^{(i)}), y^{(i)})\}_{i=1}^n$$

- **Model:** linear in $\boldsymbol{\theta}$

$$y^{(i)} \approx \Phi(\mathbf{x}^{(i)})^\top \boldsymbol{\theta}$$

- **Learning problem:** generalized linear regression

$$\boldsymbol{\theta}_\star = \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \Phi(\mathbf{X})\boldsymbol{\theta}\|_2^2 = \Phi(\mathbf{X})^\dagger \mathbf{y}$$



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Solution of Least Squares

$$\theta_* = X^+ y$$

Case II: $X \in \mathbb{R}^{n \times d}$ does NOT have full column rank

- Does not have unique solution $\min_{\theta} \frac{1}{2} \|X\theta - y\|_2^2$
- Typically, $n < \underbrace{d}_{\theta}$, thus
“**overfitting**”: more model parameters d than can be justified by the data n
- How to obtain unique and meaningful solution?

“**regularization**”: e.g., $\min_{\theta} \|X\theta - y\|_2^2 + \underbrace{\lambda \|\theta\|_2^2}_{\text{penalty}}$

where $\underbrace{\lambda > 0}_{\text{hyper parameter}}$ is the penalty parameter.

Regression and Regularization

In the regime $n < d$, **overfitting** occurs as we have **too many degree of freedom**

- **Idea:** *penalize* candidate solutions for using too many features

Regression and Regularization

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$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2$$

Regression and Regularization

In the regime $n < d$, **overfitting** occurs as we have **too many degree of freedom**

- **Idea:** *penalize* candidate solutions for using too many features
- One candidate regularization:

$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2$$

- $\lambda > 0$ is a “tuning parameter” that controls the tradeoff between fit and complexity

Regularization and Regression

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- When $\lambda = 0$
 - this is linear regression with squared loss

Regularization and Regression

$$\min_{\theta} \|\cancel{X\theta - y}\|_2^2 + \lambda \|\theta\|_2^2$$

$\lambda > 0$ is a “tuning parameter” that controls the tradeoff between fit and complexity

- When $\lambda = 0$
 - this is linear regression with squared loss
- When $\lambda \rightarrow +\infty$
 - this is minimized at $\theta = 0$

min $\|\theta\|_2^2$

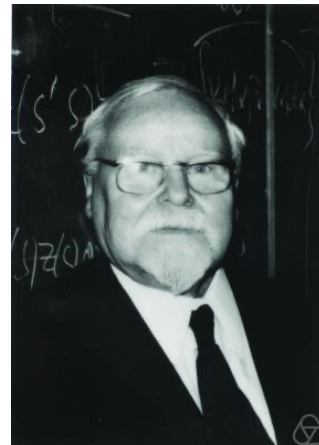
Tikhonov Regularization

This is one example of a more general technique call *Tikhonov Regularization*

$$\min_{\theta} \|X\theta - y\|_2^2 + \|\Gamma\theta\|_2^2$$

(note that λ has been replaced by the matrix Γ)

$$\Gamma = \lambda I$$



Andrey Tikhonov
Russian Mathematician

Tikhonov Regularization

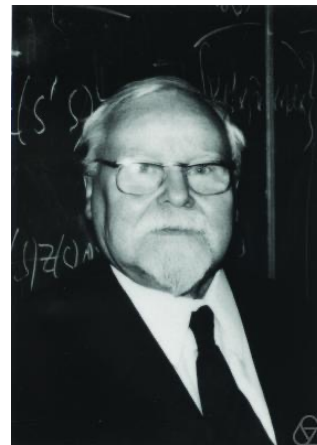
This is one example of a more general technique call *Tikhonov Regularization*

$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \|\boldsymbol{\Gamma}\boldsymbol{\theta}\|_2^2$$

(note that λ has been replaced by the matrix $\boldsymbol{\Gamma}$)

- Compute the loss function

$$\underline{L_r(\boldsymbol{\theta})} = \mathbf{y}^\top \mathbf{y} + \boldsymbol{\theta}^\top (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Gamma}^\top \boldsymbol{\Gamma}) \boldsymbol{\theta} - 2\boldsymbol{\theta}^\top \mathbf{X}^\top \mathbf{y}$$



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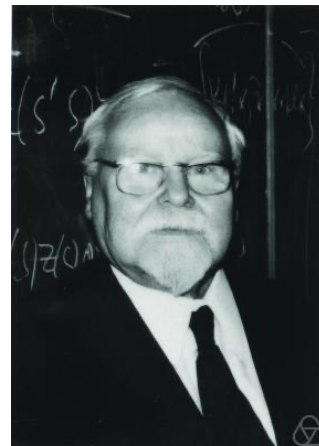
- Compute the loss function

$$L_r(\theta) = y^\top y + \theta^\top (X^\top X + \Gamma^\top \Gamma) \theta - 2\theta^\top X^\top y$$

- Compute the gradient

$$\nabla L_r(\theta) = 2(X^\top X + \Gamma^\top \Gamma) \theta - 2X^\top y = 0$$

Handwritten notes: λI (above $\Gamma^\top \Gamma$), $\Gamma = \sqrt{\lambda} I$ (to the right)



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Tikhonov Regularization

$$\min_{\theta} \|X\theta - y\|_2^2 + \|\Gamma\theta\|_2^2$$

- Set the gradient to zero: the optimal solution satisfies

$$\nabla \mathcal{J}_r(\theta) = 0 \quad \Rightarrow \quad (X^\top X + \Gamma^\top \Gamma) \theta = X^\top y$$

Tikhonov Regularization

$$\min_{\theta} \|X\theta - y\|_2^2 + \|\Gamma\theta\|_2^2 = \lambda \|\theta\|_2^2$$

(Handwritten notes: $\Gamma = \sqrt{\lambda}I$ and $\lambda \|\theta\|_2^2$)

- Set the gradient to zero: the optimal solution satisfies

$$(X^\top X + \Gamma^\top \Gamma) \theta = X^\top y$$

(Handwritten notes: $\Gamma^\top \Gamma = \lambda I$)

- Suppose $\Gamma = \sqrt{\lambda}I$, then

$$\theta_\star = (X^\top X + \lambda I)^{-1} X^\top y$$

$X^\top X + \lambda I$ is invertible for $\lambda > 0$ and arbitrary X

Ridge Regression – Closed Form Solution

When $\Gamma = \sqrt{\lambda}I$, Tikhonov Regularization has a special name: *ridge regression*

$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2$$

- It has closed-form solution:

$$\boldsymbol{\theta}_\star = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

Another Take: Constrained Minimization

- **Fact:** one can use Lagrange multipliers (KKT conditions) to show that

$$\theta_{\star} = \arg \min_{\theta} \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2$$

is formally equivalent to

$$\theta_{\star} = \arg \min_{\theta} \|y - X\theta\|_2^2$$

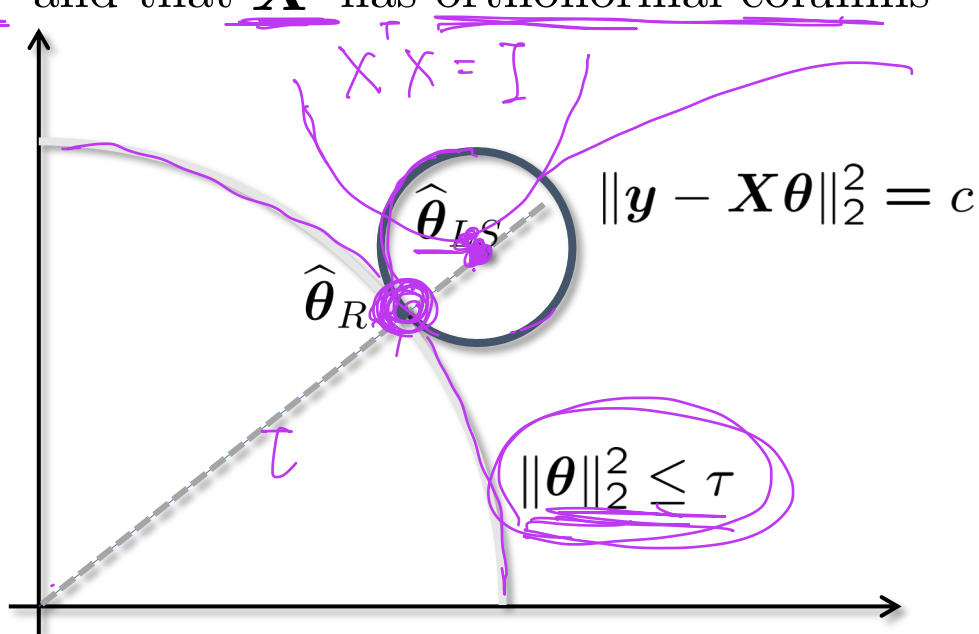
$$\text{subject to } \|\Gamma\theta\|_2^2 \leq \tau$$

$$\lambda \Leftrightarrow \tau$$

for a suitable choice of $\tau > 0$

Tikhonov Regularization vs Least Squares

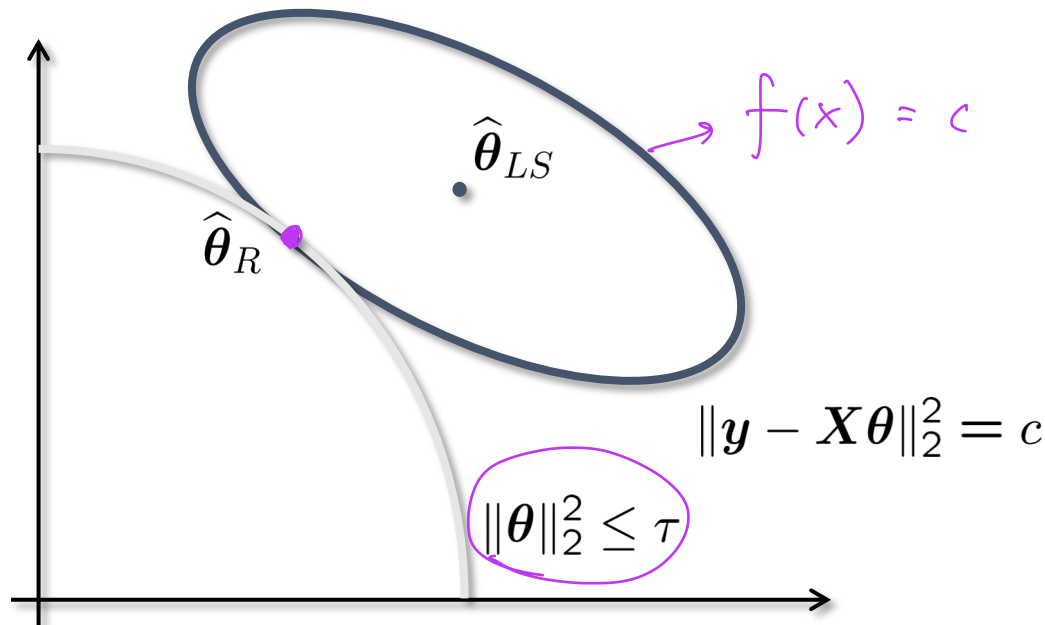
Assume $\Gamma = I$ and that X has orthonormal columns



Tikhonov regularization is equivalent to shrinking the least squares solution towards the origin

Tikhonov Regularization vs Least Squares

In general, we have the following picture



Tikhonov regularization is equivalent to shrinking the least squares solution towards the origin

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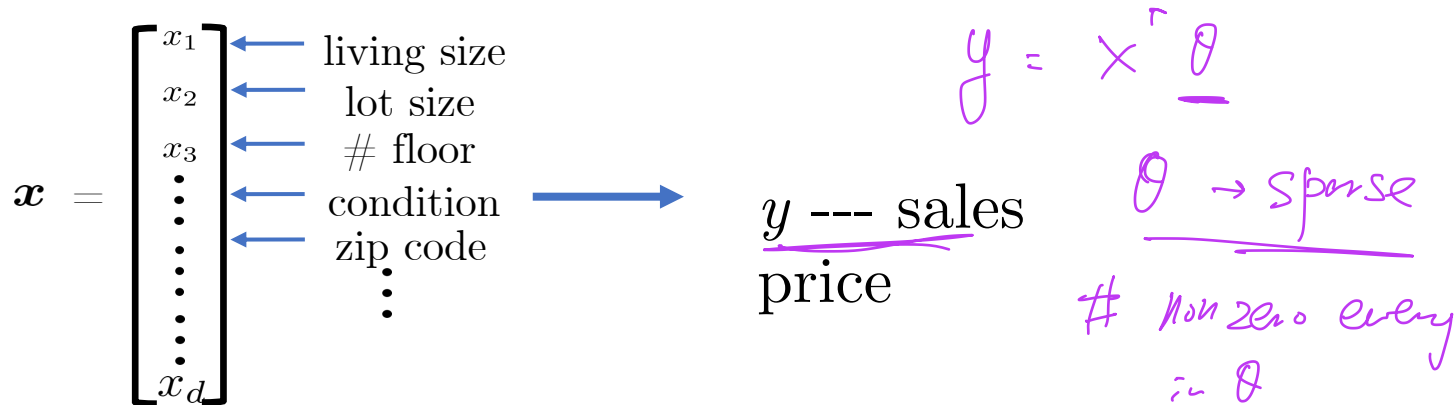
Feature Selection and LASSO

- **Motivation:**
 - When you have few examples and a large number of features $d \gg n$, it is easy to overfit

Feature Selection and LASSO

- **Motivation:**

- When you have few examples and a large number of features $d \gg n$, it is easy to overfit
- How can we remove uninformative features?



Alternative Regularizers

$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + \underline{r(\boldsymbol{\theta})}$$

- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)

$$r(\boldsymbol{\theta}) \approx \|\boldsymbol{\theta}\|_0 := |\text{supp}(\boldsymbol{\theta})|$$

d $\rightarrow$ $\exp(d)$

Alternative Regularizers

$$\min_{\boldsymbol{\theta}} \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2 + r(\boldsymbol{\theta})$$

$\lambda \|\boldsymbol{\theta}\|_1$

- Akaike information criterion (AIC)
- Bayesian information criterion (BIC)

$$r(\boldsymbol{\theta}) \approx \|\boldsymbol{\theta}\|_0 := |\text{supp}(\boldsymbol{\theta})|$$

- Least absolute shrinkage and selection operator (LASSO)

$$r(\boldsymbol{\theta}) \approx \|\boldsymbol{\theta}\|_1 := \sum_j |\theta_j|$$

- also results in shrinkage and promotes *sparsity*
- can think of $\|\boldsymbol{\theta}\|_1$ as a more computationally tractable replacement for $\|\boldsymbol{\theta}\|_0$.

The Sparsity and LASSO

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1$$

- Can also be stated in a constrained form

$$\begin{aligned} \min_{\boldsymbol{\theta}} \quad & \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \\ \text{s.t.} \quad & \|\boldsymbol{\theta}\|_1 \leq \tau \end{aligned}$$

$$\begin{aligned} \min_{\boldsymbol{\theta}} \quad & \|\boldsymbol{\theta}\|_1 \\ \text{s.t.} \quad & \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \leq \sigma \end{aligned}$$

The Sparsity and LASSO

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$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2$$

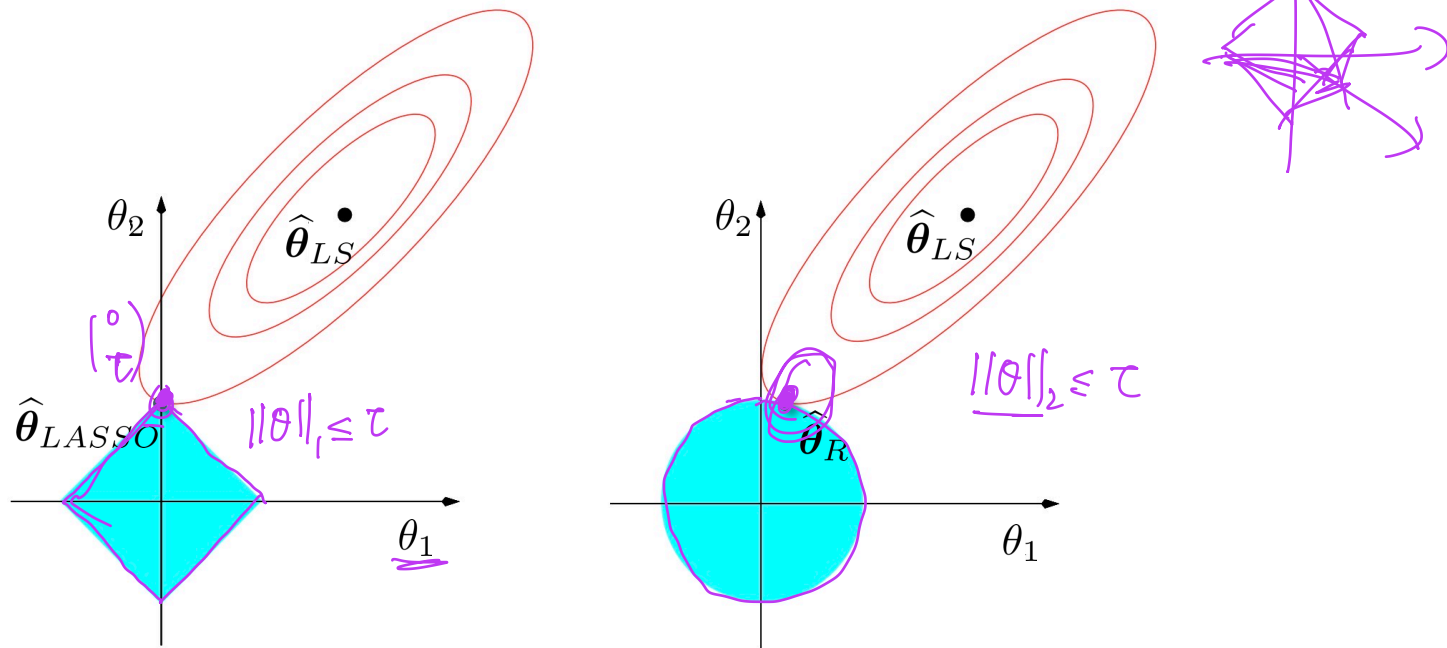
$$\text{s.t. } \|\boldsymbol{\theta}\|_1 \leq \tau$$

$$\min_{\boldsymbol{\theta}} \|\boldsymbol{\theta}\|_1$$

$$\text{s.t. } \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 \leq \sigma$$

- Tikhonov has a closed-form solution, but LASSO requires solving an optimization problem

The Sparsity and LASSO



LASSO still shrink the least squares solution towards the origin, but now in a way that promotes *sparsity*

A General Approach to Regression

Least squares, ridge regression, and the LASSO can all be viewed as particular instances of the following

$$\theta_{\star} = \arg \min_{\theta} \underbrace{L(\theta)}_{\substack{\downarrow \\ \text{data fidelity}}} + \underbrace{\lambda \cdot r(\theta)}$$

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Least squares, ridge regression, and the LASSO can all be viewed as particular instances of the following

$$\boldsymbol{\theta}_\star = \arg \min_{\boldsymbol{\theta}} \underline{L(\boldsymbol{\theta})} + \lambda \cdot r(\boldsymbol{\theta})$$

- $L(\boldsymbol{\theta})$ often called the *loss function*, enforces data fidelity

$$\underline{f_{\boldsymbol{\theta}}(\boldsymbol{x}^{(i)})} \approx \underline{y^{(i)}}$$

A General Approach to Regression

Least squares, ridge regression, and the LASSO can all be viewed as particular instances of the following

$$\boldsymbol{\theta}_\star = \arg \min_{\boldsymbol{\theta}} L(\boldsymbol{\theta}) + \lambda \cdot r(\boldsymbol{\theta})$$

- $L(\boldsymbol{\theta})$ often called the *loss function*, enforces data fidelity

$$f_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) \approx y^{(i)}$$

- $r(\boldsymbol{\theta})$ is the *regularizer* which serves to quantify the “complexity” of $\boldsymbol{\theta}$.

$$r(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_2^2, \quad r(\boldsymbol{\theta}) = \|\boldsymbol{\theta}\|_1$$

Further Readings & References

- [The Elements of Statistical Learning: Data Mining, Inference, and Prediction](#), by Trevor Hastie, Robert Tibshirani, and Jerome Friedman (Chapter 3)
- [Foundations of Machine Learning](#), by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar. (Chapter 10)